

Chapter [...]

From Evaluation to Action: Ethics, Epistemology and Extreme Technological Risk

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1. Introduction

Academic studies about how to understand and manage Extreme Technological Risks (ETR)—that is, anthropogenic and technological risks with the capacity to bring about catastrophes at the global scale²—have been taking place for many decades. However, recent research has dramatically changed the nature of this field and how it is perceived and approached. Key to this has been the emergence of a relatively unified field of Existential Risk Studies (henceforth ERS) examining existential and global catastrophic risks as a class. While there is no single moment of the field’s foundation, the publication of influential works such as John Leslie’s “The End of the World: The Science and Ethics of Human Extinction” [1996], Nick Bostrom’s “Existential risks: Analyzing human extinction scenarios and related hazards” [2002], and Martin Rees’s “Our Final Century: a scientists warning” [2003], along with the establishment of key institutions such as the Machine

¹ The three authors of this paper made quantitatively similar but qualitatively different contributions and are therefore listed in reverse alphabetical order by surname only.

² These can be contrasted with potential avenues of extinction resulting from natural events [Sandberg, 2018].

Intelligence Research Institute (2000),³ the Future of Humanity Institute (2005), The Centre for the Study of Existential Risk (2012), and The Future of Life Institute (2014), all point to a rapid development of the field over the past two decades.⁴

Of course, the past two decades have also seen a rapid growth in both the magnitude and variety of technological risks facing humanity [Deudney, 2018], from the onset of fossil-fuel-driven extreme climate change [Beard et al., 2021] to a new era of nuclear instability [Krepon 2021]; and from new synthetic biotechnologies [Noun and Chyba, 2008; Kemp et al., 2020] to the emergence of widespread, increasingly capable, and potentially misaligned Artificial Intelligence (AI) systems [Ord, 2020; Christian, 2020]. With this has also come growing awareness of-, and concern about, these threats, both amongst the global public [Global Challenges Foundation/Novus, 2020], as well as amongst policymakers at the national [House of Lords, 2021] and international levels [United Nations, 2021].

These developments point to both the opportunity and need for this field to reflect upon the nature of ERS; on how well suited this field is to its stated aim of studying and preventing threats to the future of humanity; and on how it may be further improved to that end.

We write this chapter as researchers who each entered the ERS field at different, though still early, stages in our academic careers; and did so from a range of disciplines (philosophy, the life sciences and law). We encountered challenges: in how the field was seen to function, why it appeared to function in the way that it did and what implications that had. This chapter is therefore an expanded account of how we have sought to navigate these tensions, through a process of self-reflection. In this chapter, we hope to encourage and inform such a process of

³ Until 2013 named the ‘Singularity Institute’ [Muelhauser, 2013].

⁴ For a more detailed history of this field’s development, see Beard and Torres [2020], while for its older historical antecedents see Moynihan [2020] and Torres [2021].

reflection. As such, we first provide a preliminary definition and loose sketch of the ERS community as we understand it today, before posing seven key questions facing that community, in working out the way it aims to study ETRs, to mitigate them, and in deciding its path forward. We do not provide answers, but rather, illustrations of some of the views we have encountered, with a desire to prompt further discussion and exploration.

Our goal is not to provide definitive answers to any of these questions, but rather to explore why different people might answer them in different ways. The aim is therefore to equip scholars, both those newly entering the field and those already working within well-established assumptions and practices, to take a more reflexive stance towards understanding their own research and its role in (saving) our world. While different in focus, we believe our questions all point to common challenges around direction, pluralism, epistemic modesty, diversity, inclusion, representation and accountability that the field has only recently begun to address.⁵ Given that the aspiration of ERS as a field is to ensure the future of humanity as a whole, we take these problems to be not just important but increasingly urgent; they are not peripheral after-thoughts, but pivotal ‘crucial considerations’ [Bostrom, 2014a] in charting a path forward.

The questions we ask are as follows:

1. How or why do scholars of ERS choose which extreme technological risks to prioritize and study?
2. How or why can we prioritize extreme technological risk over other global problems, and what are the possible downsides?

⁵ For an encouraging recent statement in support of diversity in methodology, ethics and approach in this field, see also the recent ‘Statement on Pluralism in the Field of Global Catastrophic Risk’ by the Global Catastrophic Risk Institute [Baum, Barrett & Fitzgerald, 2022].

3. What are the different approaches to studying and managing ETRs?
4. In what ways can the field of ERS pursue a coherent scientific approach to ETRs? (e.g. what methodologies do we rely on, and what epistemic standards should we observe?)
5. How can or should ERS scholars achieve influence and impact to mitigate ETRs?
6. How demographically diverse is the field of ERS, and what are the causes and consequences of this?
7. How can ERS best reflect on and communicate about these questions, as it continues to grow? (e.g. how can it do so while also facing up to the magnitude, importance and urgency of the problems we are trying to address?)

Our discussions of these questions should however come with an important caveat or clarification: They may read as somewhat critical of the field, or of particular groups or projects within it, including our own. In general, we think these criticisms are worthy of serious consideration and reflection, and that incisive critiques, when brought in a good-faith and nuanced manner, are an indispensable part of a healthy academic field or epistemic community. Simultaneously, we also do not wish to single out any particular group or individual as worthy of ‘blame’; the challenges we describe are systemic and the problems are hard. Our concern here is with repair rather than critique⁶ and we hope that all reading this will feel able to join in good faith efforts to make this field

⁶ In the sense of ‘critique’ discussed by Hendren [2020]. While we bring up specific critiques, therefore, we generally do not focus on the scholars they are aimed at, even if that is sometimes made explicit by the authors of these views themselves. Thus we are not being ‘critical’ in the sense of taking a negative view about people and their motives but rather the Kantian sense in which theories that are founded in an awareness of the context in which they are produced and the impact this has on the theorizer, and not in some belief about objective certainties.

even better. Mishandling our approach to ETRs would be extraordinarily bad; getting this right is too important to do anything else but learn.⁷

2. Defining Existential Risk Studies: Contours of a Community

As discussed, this chapter asks a series of questions about a community of scholarship that we refer to as Existential Risk Studies, and the way that this community seeks to understand and mitigate Extreme Technological Risks. While we believe that there are good empirical and normative reasons to view ERS as an emerging academic discipline in its own right, it is worth noting that it is still not widely recognized as such. In particular, many of the usual hallmarks of such a distinct discipline—such as named academic departments, specialized technical journals, or formal academic training programmes—are either non-existent or still at a very early stage of development.⁸ We must therefore begin by saying a little about how we understand both the topography of this community, the shape of the field—and our own relationship to both.

The notion of a community of ERS that we use in this chapter is that of a community of alignment, representing those researchers who are actively seeking (1) to align their research so that it makes the greatest

⁷ ‘Sayre’s law’ states that “in any dispute the intensity of feeling is inversely proportional to the value of the issues at stake [...] That is why academic politics are so bitter.” [Issawi, 1973:178]. Given that most involved in the ERS are genuinely motivated by the very high importance of addressing ETRs well, we expect and hope that critical debate on this important topic can prove the corollary of this law, and strive to remain nuanced, modest and good-faith—exactly because the stakes are so high.

⁸ Although there are, notably, an increasing number of initiatives aimed at supporting and cultivating young research talent in this space, such as the Stanford Existential Risk Initiative (SERI), Cambridge Existential Risk Initiative (CERI), and a diverse set of research fellowships and programs.

contribution to our understanding and mitigation of ETRs and (2) to align their research with others who share the same aim so as to create the best collective results.

It is likely that this definition could be applied to multiple different communities within ERS, including those who share different understandings of ETRs.⁹ We therefore note that the community we are addressing is one which we ourselves are most closely aligned with—that of an institution (CSER) explicitly studying ETRs. We do not view this community as inherently exclusionary; as any researcher belongs to it by dint of simply seeking to align their work with other members of the community. However, as we will describe, in practice there are ways in which the field may appear somewhat exclusive or unintelligible to researchers coming in from other academic disciplines. As such, there is a significant gap between (1) the groups we identify as forming the core community of ERS, and (2) the much wider community of all researchers whose work is or could be relevant to the understanding and mitigation of ETRs (but who are not directly seeking to align their work to these aims or with the work of others in the field), and (3) the still wider community of everyone (not just academics or researchers) concerned with understanding and mitigating ETRs.

With that in mind, it is also worth acknowledging that as authors we stand in at least three relationships with the field of ERS. Firstly, it is the subject of this chapter and of our research and we are writing based on our understanding both as insiders who have direct experience of this community, and also as outsiders who bring distinct tools and expertise from our previous disciplines (philosophy, pathology, and law) to bear in studying the field and the community. Secondly, it is also the field that we address with this chapter and one that we are looking to positively influence. And finally, it is a field that we self-identify with and believe

⁹ For a discussion of different principal approaches or clusters in the ERS field, see our discussion of Question 3.

ourselves to be part of. We wish to stress that our view is that it is the same community we stand in these three relationships with.

A second consideration relates to our understanding of Extreme Technological Risks. This is itself a very interesting philosophical question that has received quite a bit of attention within the community, and we do not wish to rehash any of those arguments. Instead we will simply note that we do not believe that it is necessary for a risk to threaten immediate extinction in order for it to count as ‘extreme’: rather, any risk with an impact that could be global in scale, that might alter the future trajectory of the human species in irreversible ways—including through severe harm to well-being or values—that lies beyond the scope of existing risk management mechanism, would all be extreme. Similarly, we do not believe that it is necessary for a risk to emerge from *new* (or future) technologies in order for it to be classed as ‘technological’: a technological risk could be one that emerges from technologies that have existed for centuries (like burning fossil fuels); or even one that stems from a failure to develop technologies that might otherwise have saved us from other extreme risks (like planetary defence programs, or geoengineering).

Question 1) How or why do scholars of ERS choose which extreme technological risks to prioritize and study?

A given technological risk can be defined as an *extreme* technological risk (ETRs) on the basis of their magnitude (scope/severity), their (ir)reversibility and permanence [Bostrom and Cirkovic, 2011], or their relation to critical global systems on which humanity depends. Regardless of how they are defined however, the world is currently facing many such extreme risks, with numerous critical system failures, global spread mechanisms, and mitigation fragilities combining to imperil our species [Avin et al., 2018]. There are clearly many potential

problems, and a wide range of topics that could need studying. How do scholars select or prioritize topics to study?

Of course, at a straightforward level, any academic's choice of research topics or domains is often the result of an iterative combination of many factors. This includes both 'internal' (or object-level) factors, such as researchers' personal convictions about the world's problems, and judgments about which to prioritise given a certain heuristics or principles for 'cause prioritization'. It also includes 'external' factors of certain research lines, such as over how they might affect career security or capital (e.g. likelihood of publication, perception of how it will affect future funding or career prospects), as well as other influences on decision-making (such as cognitive biases; perceptions of social capital or esteem in the larger epistemic community of ERS).

How do such dynamics feature into and affect the research portfolio in ERS? Interestingly, while there is a wide range of potential ETRs, the work of 'core' ERS scholars often seems to cluster around a subset of particular issues: nuclear weapons risk, biosecurity, AI alignment and, to a lesser extent, food security and social and ecological collapse.

A number of reasons have been given for the constitution of this portfolio or research agenda in the community. Below, we will critically consider three different factors, and how they may contribute to or shape research agendas and priorities in ERS in ways that may be both good and bad. These are (1) 'cause prioritization' heuristics employed by the community; (2) how the disciplinary background of researchers, and its historical genealogy, may shape research agendas; (3) how considerations of research funding may affect research priorities.

Explicit cause prioritization frameworks & principles

To start with personal researcher's cause prioritization: there are of course many approaches to prioritizing issues, ranging from pure intellectual interest to perceived overwhelming societal importance.

However, one that has been particularly visible in- and characteristic to ERS, has been the development of a particular model of ‘cause prioritization’, called the ‘Importance, Tractability and Neglectedness’ (ITN) framework.¹⁰ Under the ITN cause prioritisation framework, many ERS scholars have come to study particular risks through an analysis that considers not just the *importance* or significance (considered as a product of the number of people affected by a risk, the magnitude of that risk’s impact and the risk’s certainty or probability), but also the *tractability* of mitigating that risk, and the *neglectedness* of the problem in terms of resources and attention that are allocated to it today (as well as which are likely to be allocated to it in the future, before it is too late).¹¹

As such, under the ITN model, while any existential risk is *prima facie* important (i.e. affecting huge numbers of people, and large parts of the future, in extreme ways),¹² the size of impact alone is not enough to warrant prioritization. Rather, the additional consideration of ‘tractability’ recommends that, all else equal, people prioritize work on risks where they can expect to make a meaningful contribution to solving that problem—given both the features of the problem (e.g. whether or not it is deeply politicized) as well as the resources or expertise available to

¹⁰ See Wiblin [2016], and MacAskill [2018]. For one critique of the framework, see Dickens [2016].

¹¹ Winter et al. [2021:23-24] provide a definition of the ITN framework, as one where “Importance refers to the number of sentient beings affected and the degree to which they are affected by a given problem. To put it simply: “If we solved this problem, by how much would the world become a better place?” Tractability refers to the possibility of actually solving the problem. As a heuristic, one might ask: “If we doubled direct effort on this problem, what fraction of the remaining problem would we expect to solve?” Finally, neglectedness refers to the question: “How many resources will be dedicated to solving the problem before it is too late?”.

¹² Though notably, this does not always, for all scholars, extend to global catastrophic risks which might have an immense impact on today’s world, but which could allow for eventual civilizational recovery [e.g. Rodriguez, 2020]; such risks are considered important, but not as important as risks of permanent extinction, which are held to foreclose a potentially deep future.

them (e.g. particular types of academic expertise, or particular networks). In addition, the ITN framework also emphasizes ‘neglectedness’, recommending that (all else equal) people steer away from working on risks where very many others are already seeking to make such contributions, and rather emphasizes work on relatively neglected risks, to ensure that the marginal impact of the novel or additional attention may be relatively large.

Under this ITN framework, we can find one explanation for which ETR topics or hazards are most-studied in the ERS field. In this argument, the existing issue portfolio (with an emphasis on issues such as AI risk or nuclear risk) simply reflects the underlying strategic landscape facing many in the discipline: ERS scholars (should) focus on the most significant problems, that are the least addressed, where they can personally expect to make the biggest marginal difference, given their expertise, career capital and fit.

In a world with limited resources and many uncertain challenges to be addressed in parallel, there is an underappreciated value to the importance-tractability-neglectedness cause prioritization approach to cause prioritization. However, while attractive and innovative, there remain a number of potential problems with using this cause prioritization framework as a way to estimate the size of marginal impact that scholars can expect to have on a problem or ETR.

Firstly, some of the ITN terms may be very hard to assess *ex ante*. While there are situations where we could base ballpark estimates for the full costs of successful ETR mitigation interventions on prior information, there may also be at least some kinds of low-information environment ETRs where it is not until a problem has been solved that we can actually know what amount of effort it took to solve it. Indeed, ERS researchers have pointed out that crises such as the COVID-19 pandemic illustrate domains where the ITN framework may not perform well, or will need to be modified to be fully appropriate. For instance,

Kulveit & Leech [2022] have suggested that ‘static’¹³ versions of the ITN framework perform well in situations where (1) we are allocating marginal resources seeking marginal impact (e.g. individual ERS researchers deciding what risks to study), (2) impact estimates are not changing rapidly, and (3) the whole action space is understood at the moment of prioritization; they suggests that these assumptions are violated in crisis situations such as the COVID-19 pandemic, which (at least in its early stages) created a situation where it was “hard to estimate the importance, tractability, or neglectedness of different actions; the future options in action space were unknown, and the impact of a given actor was highly contextual. Also, it was possible to get into a position to influence the total allocation of resources” [ibid]. As such, some ETR domains may offer limited initial knowledge, and/or an opportunity to allocate not just marginal resources (e.g. researcher attention) but rather a large fraction of all resources (e.g. research programs), or be rapidly evolving. In those cases, conventional ITN approaches may fall short, and there may be a need for a modified, dynamic approach to ITN prioritization,¹⁴ which puts more emphasis on exploring the potential action space, and on increasing future optionality.

Secondly, assessments of tractability and neglectedness are very sensitive to how a problem is framed. For instance, some ERS scholars view climate change as a relatively non-neglected field, given the size of the climate science field as a whole, and its global public attention [e.g. Todd 2017]. *Within* climate change research however, the potential for climate change to be an existential or global catastrophic risk *is* in fact very neglected, with the focus squarely on higher-probability but

¹³ Or ‘argmax’--that is, an approach that only evaluates a relatively small number of options; where their importance, neglectedness and tractability are estimated from expert judgment as point estimates, and where only the top options are allocated resources [Kulveit & Leech, 2022].

¹⁴ i.e. ‘softmax’ prioritization, which gives a distribution of options weighted by how likely they are to be the most impactful.

lower-impact climate trajectories [Beard et al., 2021; Jehn et al., 2021]. As a result, some scholars in ERS have emphasized the importance of developing the ITN approach in order to accommodate ‘cross-cause prioritization’, which is capable of considering not just the overall neglectedness of one ETR (e.g. climate change) relative to another (e.g. AI risk), but rather can also account for the *relative* neglectedness of that particular ETR within particular high-leverage disciplines such as legal scholarship [Winter et al., 2021: 100-101].

A third problem for the ITN model of cause prioritization within ERS is the risk of minimizing the influence of cognitive biases on assessments of the three terms. Given given deep epistemic uncertainty,¹⁵ and given different moral intuitions as well as differences in scope (in-)sensitivity around ETRs, there may be a lot of room for individual and group biases to affect which problems are seen as priorities, and thus to be perpetuated within research without sufficient challenge.¹⁶ To be sure, the emphasis on ‘cause neutrality’ [Schubert, 2017] is admirable, and most approaches to cause prioritization explicitly value the importance of constantly re-evaluating existing causes, as well as remaining on the lookout for potentially unexplored yet critical causes—a so-called ‘Cause X’ [MacAskill 2016]. Yet while such commitments to exploration and re-assessment may mitigate such biases somewhat in theory, they may not always work effectively in practice; or where they do, they may not suffice to overcome the challenges entirely, resulting in only incremental re-adjustments to approaches to pre-existing domains.

¹⁵ See also Sundaram and Currie [2022 forthcoming], in this volume.

¹⁶ Of course, that is not to say that such framing problems only affect this three-part cause prioritization framework: indeed, more conventional prioritization approaches which focus only on a problem’s ‘overall importance’ might be nominally immune from biases colouring their assessment of tractability and neglectedness, but might be all the more susceptible to biases that affect their perception of an issue’s ‘importance’.

A different interpretation of the standard claim about ITN cause prioritization within ERS suggests that the incorporation of assessments of importance, tractability, and neglectedness help form a kind of ‘epistemic democracy’, in which the community can debate possible priorities and work out a reasonable consensus or reflective equilibrium about how to distribute effort. Assuming that its members have access to at least some useful information about how to prioritise their research, that they can reason independently, and that power is somewhat evenly distributed across the community, such a democracy could be expected to arrive at better decisions than any individual researcher deliberating on their own. Unfortunately, all of these conditions can be brought into doubt. For instance, given the unprecedented nature of existential risks it is entirely possible that researchers do not have access to useful information about them (in terms popular within the community, it is possible that everything is pure ‘noise’ and we have no good signal about the risks that we face) [Beard et al., 2020; Baum 2020; Morgan 2014].

Moreover, ERS scholars should reckon with the risk that they might experience a high degree of interdependence, not only about the information they can access, but also how they reason about it, given shared background assumptions about epistemology, (meta-)ethics, and political processes in the world. While the community puts considerable emphasis on epistemic rigour and caution, it still should not expect to be immune to general social community problems such as selection bias and motivated reasoning.¹⁷ While many of these features would hardly make the field of ERS unique amongst other academic fields, communities, or movements, these are still features that would tend to undermine its ability to realize the full value of epistemic democracy; at the very least, they point towards avenues for improvement.

¹⁷ This case has been made more generally (and reflexively) about the Effective Altruism movement [Zhang 2021], and also might apply to the longtermist and ETR-focused sub-communities within it.

In briefly raising these critiques we do not mean to dismiss the idea that standard views about cause prioritization within the ERS community are wrong. However, we do think that they give us reason to also consider more critical alternative theories that some have advanced.

For instance, Kuhlemann [2019] argues that the community is overly concerned about problems that share three features in common. First, they are *epistemically neat*, making it easy to identify which disciplines are best-suited for studying them. Second, they have a *sudden onset* in that they “crystallise abruptly, with obviously catastrophic outcomes from as little as a few hours to, at most, a few short years...”. Third, they are *technologically driven* and, as such, “have a close relationship with rather flattering ideas about human ingenuity and intellectual prowess” [2019:8]. Kuhlemann characterizes risks with these three properties as “sexy”, and contrasts them to the many “unsexy” problems that are complex, slow moving, and relate to more fundamental issues with human society or organization—such as issues of overpopulation and environmental breakdown—and which are thus less ‘interesting’, even if they may be no less important. At the same time, while this may have held for parts of the community in its early years, recent years appear to have seen a greater exploration by ERS scholars of complex and systems approaches to ETRs [Liu et al. 2018].

Admittedly, concerns of excessive or convenient ‘epistemic neatness’ could be raised not just against overly linear models of ETRs, but to many approaches in ERS. For instance, theories that focus primarily on which actors are to be held responsible for the creation of ETRs can certainly provide important (and still-underexplored) insights to ERS. At the same time, any analytical lens that focuses on ‘who is to blame’ may need to be careful to avoid the common human tendency towards moralizing psychology [e.g. Ballentyne and Ditto 2021]. Such tendencies could make accounts focused on deliberate choices and culpability more ‘epistemically neat’ than more messy perspectives that understand ETRs as the emergent outcome of various unintended,

multi-actor or incentive-driven systemic drivers, yet that need not mean that they are always more accurate. Indeed, scholarship on crisis management have noted that accountability processes can be a double-edged sword [Boin et al. 2016]: such mechanisms are on the one hand indispensable to bringing public closure to crises; yet the politics of accountability can also undermine institutional capacity to learn, as the escalation of ‘blame games’ may gridlock accurate assessments of failure, as information and data become pressed into competing narratives deployed in battles for personal and institutional survival [Boin et al. 2016: 131-132]. Finding a way to balance the avoidance of such derailing outcomes, achieving institutional learning, while still ensuring accountability, may be a key challenge for ERS.

Similarly, accounts that neatly fit onto pre-existing partisan narratives may be more epistemically salient, but could result in a convenient dovetailing of our preferred ETR mitigation strategies with our already-previously-held political programs or ethical preferences. Such a ‘suspicious convergence’ [Lewis 2016] need not be grounds to rule out our preferred strategies, but it might nonetheless be a good reason to critically examine our reasoning. These clearly are complex debates that will require much further exploration, and even then will not yield hard-and-fast rules.

Background disciplinary history & assumptions

Beyond explicit heuristics or principles for prioritising ETRs, what other influences shape ERS research agendas? As with many other academic fields and disciplines, the selection of areas of priority, and of which methods to use in ERS, might also reflect underlying assumptions that can be traced back to disciplinary paradigms, blind-spots and worldviews. These may include assumptions that modern academics or the public might question or critique. ERS could benefit from discussion on such issues. For instance, recent years have seen some raise more acute critical perspectives on ERS, such as Mitchell and Chowdhury’s

[2020] argument that the field, like many in academia, is pervaded by ideas that protect and advance a view of humanity that is embedded in eurocentrism and white supremacy, even if those who work in the field may not be aware of this.

Crucially, such genealogical arguments do not at all presuppose that researchers working in this field are knowingly or intentionally pushing a racist agenda; rather they argue that there are certain assumptions buried deep within our ways of thinking about the world, which reflect or bear traces of a worldview of those who did engage in racist projects. For instance, Mitchell and Chowdhury highlight how “many authors working in this genre worry about the interruption of the perceived stadial progression of ‘humanity’”. This is an idea with roots in 18th century moral philosophy and economics, where it was explicitly linked with the supposed superiority of ‘modern’ European societies and the justification of their colonial expansion (albeit with a greater or lesser degree of wider humanitarian concern). Modern thinkers may feel that this connection no longer stands, or that key ideas such as the ideal of historical moral and economic progress can and perhaps should be separated from such negative associations and historical abuses.

However, the risk remains that such clearing is only superficial, and overlooks questionable assumptions that continue to reflect this historical origin. This could include a range of views that persist across many academic disciplines, and similarly could play a role in ERS. These might include, for instance: the notion that societies can be evaluated ‘on the whole’; that the terms of evaluation can be adequately set by members of the supposed ‘advanced’ societies; that progress can be associated with simple economic growth; that the direction of historical travel is linear and unidirectional; that it would be on the whole better if all societies adopted the socio-economic model of the most ‘advanced’ rather than allowing for a plurality of structures, and so on. Any of these—while of course having important implications about the

composition of the field—also likely impact how researchers in the field prioritize.

That is not to say that all of these tenets prevail across all (or any) parts of ERS, but the field should still encourage careful and nuanced debate. Encouragingly, there are indications that the ERS community has begun to grapple with these debates over how it relates to, learns from, or can correct for, the historical roots of certain ideals, or the past misuses or unintended consequences of certain approaches and assumptions (noting that these ideals, approaches or assumptions have been far from universally-held).¹⁸ This may be a long discussion, but one that is overdue. Again, we don't intend to imply that these critiques are all correct. It remains, however, the case that the field in its current form remains largely white, male, and privileged, as well as dependent upon philanthropic funding from wealthy individuals [deluks917, 2020]¹⁹ the implications of which we discuss below in Question 6. Moreover, while the ERS community has been eager to draw from many new disciplinary lenses,²⁰ it has shown limited interest in certain alternative perspectives, such as those from heterodox economics, intersectional feminism, deep ecology, or BIPOC (Black, Indigenous, and People of Colour) futurisms. Of course, it might be argued that this is not a unique feature of the ERS community, and rather reflects a more general split that appears to run through many academic fields, between 'mainstream' and 'critical' approaches. Nonetheless, it may be plausible that some of these alternative fields could offer interesting new interventions or perspectives on key questions and crucial considerations for the ERS field,²¹ even if (or exactly because) they represent distinct ethical and

¹⁸ See for example Manheim [2021b].

¹⁹ As discussed shortly.

²⁰ Such as many fields from across the social sciences [Irving and Askill, 2019], law [Winter et al., 2021], or the humanities [Vaintrob, 2021].

²¹ To give some potential examples (which would require much further elaboration): debates over how ERS scholars should view the macro-strategic

intellectual traditions to those common in ERS. That is not to say that their proposals are necessarily without shortfalls, or to be uncritically accepted.²² However, entering into conversation with this broader range of academic disciplines remains critical not just from the perspective of potentially finding new solutions or ETR-mitigation strategies, but also in light of building a broader academic coalition or community to support ETRs as important problem to be addressed. Either way, taking a critical perspective about our study of ERS can help to highlight both the potential downsides of, and also interrelations between, these features of our work. Importantly, this could also enhance more reflective prioritization efforts for the field as a whole.

Career considerations & funding landscape

A third consideration driving prioritization in ERS, may simply have to do with pragmatic considerations around career runway and funding opportunities. Of course, in any scientific field, the focus area of research (whether considering particular sources of risk, or a particular method of understanding risk) will partly and necessarily be shaped by the availability of funding to undertake that research.²³ It is relevant to

and ethical implications of technological interventions to mitigating or stabilizing ecosystem collapse ('techno-adaptation'), could draw important insights from the fields of 'green transhumanism' [Holt, 2021]; arguments over the value of the long-term future can draw on complementary ecocentric arguments for space expansion ('greening the universe') [Owe, 2022]. Similarly, the discipline of heterodox economics draws on models of steady-state economics: if these could be shown to be sufficiently stable over longer-term timelines, such strategies could offer complementary or alternate avenue to long-term civilizational wellbeing, that could be more reliable and tractable than accounts that pursue 'sustainability as trajectory' [Bostrom 2013].

²² For example, see the recent argument by Peutherer [2021], responding to suggestions that a long-termist concern for the far future necessarily mandates a rejection of capitalism, by arguing that in fact many ETR challenges could still be addressed by more limited reforms within that system.

²³ This may be mitigated slightly in the ERS field, relative to others, by the fact that recent years have seen the emergence of a relatively large community of

consider here the surprisingly relatively narrow and idiosyncratic funding base of some work in the ERS field, which has in recent decades drawn to a notable extent from donations from wealthy philanthropists, often from the tech sector.²⁴ With the emergence of new initiatives such as the FTX Future Fund [Beckstead et al., 2022], this trend seems set into the future, potentially offering much-needed funding for the field.

Of course, the primacy of philanthropic funding may have benefits, and might be necessary (or even for the best) in ERS: research suggests that standard academic funding bodies are generally conservative and cautious, prefer work with clear, direct and near-term applications, and may be ill-suited for considering truly interdisciplinary work on highly uncertain topics [Currie, 2019]. By contrast, philanthropic models of research funding, at their best, are able to pursue a more explorative or ‘hits-based’ [Karnofsky, 2016] approach to funding early-stage work, landscape assessments, or interventions on problems with high uncertainties. This could make funding models offered by organizations such as the Open Philanthropy Project or the FTX Future Fund unusually (or at least comparatively) well-suited for supporting work on many ETRs, where there may be only limited historical precedent or base-rates to learn from—or to be motivated into action by, and where success may require the funding of projects that are uncertain but have high expected value, in that they could have a tremendous positive impact, even if their probability of success is low [FTX Future Fund, 2022a]. Indeed, the ability of philanthropic models to be able to initially fund and pursue not just analysis of ETRs but also novel interventions on ETRs might be especially worthwhile, as doing so could provide early evidence on these interventions’ (in)effectiveness,

junior individuals highly motivated to undertake research, even in non-academic routes or funding arrangements. However, as also discussed further (in Question 6), this trend may have its own challenges around diversity.

²⁴ See for instance Cremer and Kemp [2021:1] for an indicative discussion of prominent funding sources in the ERS field.

and so can inform future investment decisions on ETR mitigation [Askill 2019].²⁵

Nonetheless, such philanthropic sources of funding also present challenges: these include (1) the potential shaping of research agendas in ways that may be insular or at least miss crucial considerations; (2) potential challenges about the inability to easily fund certain interventions (such as intergovernmental institutions such as the under-funded Biological Weapons Convention²⁶) which may either gain their legitimacy from public funding, or where there are regulatory funding limits on private donations (e.g. around electoral campaigns or lobbying); they may (3) be vulnerable to sudden downside shocks that would simultaneously affect large parts of the funding pool (e.g. a future tech sector or crypto -bubble); (4) the sustained efficacy of the interventions may risk putting ‘all eggs in one basket’, standing or falling with the continued accurate judgment of a relatively small group of grantors (or re-grantors), who may simply not be able to sufficiently keep track of relevant developments and scientific insights in all the relevant fields to assess highly variable interventions or grants; finally, (5) the philanthropic model could also foreclose more foundational critiques of or to the ERS field.²⁷ For instance, a deep critique of such funding

²⁵ On the other hand, it is important to keep in mind that a funding or resource allocation portfolio that emphasizes exploring more uncertain interventions over exploiting more established or proven routes may also have drawbacks, even or especially in the domain of ETR mitigation. Furthermore, even if ‘hits-based giving’ is an uniquely suitable approach valuable to ERS, that need not be the case that only elite philanthropy is ever able or willing to provide such funding. Indeed, it could be an important goal for ERS to underscore the importance of hits-based-giving approaches generally, and ETR mitigation interventions in particular, to a wide range of stakeholders or funders.

²⁶ Though there have certainly been calls to do so, see Walker and Koblenz, [2017].

²⁷ Though it should be noted that, in order to alleviate such risks, recent initiatives such as the FTX Future Fund have emphasized the importance of funding potential critiques, noting that “We’re very excited about growing the effective altruism community. And we want to critically examine its ideas, to

arrangement could draw from Anand Giridharadas's [2018] critique of how elite funding of work that (often very sincerely) purports to be "world changing", could nevertheless end up propping up a worldview that is sympathetic to and supportive of certain capitalist interests, and dismissive of the potential need for fundamental systemic change, or the redistribution of power and privilege.²⁸

Of course, that does not mean that elite philanthropy is a uniquely poor source of funding for important ERS research. Nor that these tradeoffs are unique to it. Indeed, many other sources of funding will likely present similar balances of benefits and costs that will need to be navigated: *traditional academic research council* funding may often be too cautious and epistemically conservative, as discussed. *Commercial funding* can introduce potential conflicts of interests and (real or perceived) threats to academic integrity [Abdalla and Abdalla, 2021]; the same risks might hold for *government funding*, given the intimate implications that many ETR conclusions may have for national policies or interests,²⁹ or the risk that ERS scholars become perceived as aligned to certain national interests, when effective risk mitigation on some domains may critically require that the community continues to be perceived as an 'honest broker' [Pielke, 2007] in these domains. Civil society action may be important for risk mitigation, but is rarely a direct source for funding of academic research: even if they did provide such funding, they may often (legitimately or not) remain focused on more urgent and visceral issues. For instance, there are certain areas and ETR topics where conventional 'issue networks' have at times proven initially

help ensure that effective altruists cultivate open-mindedness, resist insular thinking, and avoid homogeneity" [FTX Future Fund, 2022b].

²⁸ However, for a recent response to such 'hyper-criticism' of philanthropy, see also [Breeze, 2021], arguing that while not perfect, philanthropy continues to offer significant value distinct from government and market.

²⁹ Including in ways that sometimes reflect well on states, but on other occasions may reflect more critical perspectives on the ways they might either fail to address, or even contribute to various ETRs [Kemp, 2021].

uninterested in issues that are seen as ‘too speculative’: such was arguably the fate of ‘killer robots’ which, while eventually the subject of a concerted public campaign, were initially perceived as too futuristic by humanitarian disarmament NGOs which served as gate-keepers to key international fora.³⁰ In such cases, there may be a necessary role for funding models that will support early or critical timely attention for (potential) ETRs.

Ultimately, there may be practical and instrumental (as well as normative) reasons why it would be valuable for ERS to find ways to diversify its funding source portfolio, both across the above funding categories, but also potentially beyond it, such as drawing from broader social movements.³¹ This could enable some of the above constraints and challenges in agenda-setting to be overcome or at least balanced. At the very least, the ability to be reflexive about sources of funding, their impacts and influence (actual and perceived) on research agenda, and transparency about them to researchers and public, will be of high value as the field matures.

All this is not to say that any of these three factors—(1) individual’s explicit cause prioritization principles; (2) background disciplinary or methodological influences; (3) the funding landscape—are illegitimate. However, we think it is key for scholars in ERS to both recognize the existence and effects of all three factors, and to open up more reflexive debates about the implications of these factors. That can help support a more critical self-reflection by both individual researchers and the community, on how choices made on all three fronts shape their research agenda—and whether, or how, these may contribute to the mitigation of ETRs, or when they may undercut that project.

³⁰ See for instance Carpenter [2014], who chronicles early attempts in the early 2000s’ to put military AI technology on the global human security agenda, and how these ran into indifference or even dismissal as far too futuristic.

³¹ As indeed the Effective Altruism movement has managed to some extent.

Question 2) How or why can we prioritize extreme technological risk over other global problems, and what are the possible downsides?

As should be clear from the above discussion, any field will prioritize some issues over others; in some sense, this may be core to its identity as a ‘field’. Yet how can we compare not just amongst different ETRs; but also amongst ETR mitigation and important other challenges in the world?

As Max Roser has said in the context of development studies: “The world is much better, the world is awful; the world can be much better” [Roser, 2021]. The final part of that statement is what seems to motivate a great deal of the overarching research climate today, which has increasingly focused on solving issues of global concern. Indeed, this philosophy may be formalized in various ways that include funding regimes such as UKRI’s Global Challenges Research Fund, the Gates Foundation’s concept of ‘Grand Challenges’, or broader organizing principles and tools such as the UN’s Sustainable Development Goals [Schot and Steinmueller 2018; Flink and Kaldewey 2018]. Many well-established and pressing global problems, from sustainable development to education, from global health to international peace, remain poorly or insufficiently understood. Even if there are already many talented scholars working on these issues, why should we justify diverting any resources and talent to focus on highly uncertain future risks, when these concrete challenges remain unsolved and are facing us *now*?

Dissolving tradeoffs between ETR mitigation and other global problems

To be sure, while ERS often claims the importance and urgency

of problems in the field,³² it would not argue that these present-day issues are not important. Still, how can we ground an argument in favour of focusing attention on ETRs?

A first answer is to challenge the description of ERS as only focused on highly uncertain or speculative risks. As clarified in our earlier definition, ETRs are not only about future technological risks, but also extend to risks building up from long-established and deployed technological infrastructures, such as nuclear weapons or fossil fuel technology. Moreover, scholars in ERS have in recent years argued a ‘common-sense case’ [Wiblin and Harris, 2021] for mitigating existential risks, on the argument that, far from being extremely unlikely and far away, many may have terrifyingly high annual probabilities, accumulating to significant probabilities over our lifetimes, and that accordingly their mitigation would pass a simple cost-benefit analysis.

A second answer is to challenge the binary between ‘near-term’ and ‘long-term’ problems³³ in a more fundamental way. Exploring both (types of) issues in light of one another can help us with important choices over where to ‘steer’ the world [Karnofsky, 2021b]. In particular, examining uncertain but plausibly terminal future trajectories can help provide additional rationales for re-directing current societal trajectories: to make the case that business-as-usual is truly not sufficient, not only today, but also in a longer-term perspective.³⁴ Indeed many of the current challenges facing the world are themselves ETRs, or are closely related

³² Along with, on some accounts within the ITN framework, the relative tractability and neglectedness of those problems over present-day ones, especially those that are highly politicized.

³³ See also MacKenzie [2020], arguing that ‘there is no such thing as a short-term issue’; and [Prunkl and Whittlestone, 2020; Stix and Maas, 2021] for this argument in the domain of AI ethics.

³⁴ In doing so, this macro-strategic work into the implications of ETRs can also provide key considerations that reframe what aspects of today’s concrete problems we seek to address in the first place, or in what order. In that sense, such work can be ‘problem-finding’ [Liu and Maas, 2021].

to them, and should therefore already be considered important and legitimate issues to study for those concerned about ERS (even if there are also already communities studying them from a non-ERS perspective). Work on concrete and urgent challenges today speaks to the fact that ETRs are not just a thing of the future but already of the past and present.

From algorithmic surveillance to the climate crisis, it is often the case that, at least to begin with, the negative impacts of technology fall earliest and most disproportionately on those who are worst off and most marginalized. These negative effects can take the form of death and physical destruction, but can also involve more intangible irreversible losses such as the loss of cultures or languages. While presently occurring at scales far lower than those of an existential or global catastrophic risk, such costs can be seen as the canaries in the coalmine, highlighting both how current deficits in governance indicate likely deficits in managing and mitigating more extreme risks,³⁵ but also how rapidly developing technologies are already very poorly aligned with human values, which could give us reason to suspect that their further development may not radically or suddenly dissolve this problem, but instead scale up their harms even further, barring urgent action to curb or re-direct these trends already today.³⁶

Third, there is the issue of co-benefits: interventions that can

³⁵ See for instance Liu et al. [2020] and Krakovna [2020], both describing early lessons from societal failures in responding to the COVID pandemic, for other ETRs.

³⁶ Interestingly, some scholars in ERS have expressed concern over the potential of ‘value-lock-in’—the scenario that “we are coming to a point in time where we will invent a technology that will enable the agents alive at that time to maintain their values indefinitely into the future, controlling the broad sweep of the entire rest of the future of civilisation.” [MacAskill, 2020]. Yet, as Cremer and Kemp have memorably suggested, ‘if the world is locked into the development of dangerous technologies, then we are already in a “lock-in scenario”, with the majority of the world currently having little control over whether certain ETRs are pursued further.

help us address existing challenges can often help us prevent, mitigate, or recover from future catastrophes, and vice versa. Maintaining robust and resilient global systems, redressing injustice and inequality, reinforcing norms around shared values, improving governmental risk and resilience strategies,³⁷ and improving international cooperation are all goals that can contribute to a better world today; but they also reduce the likelihood of severe harm from ETRs down the line. Poverty, ill health, and ignorance are major sources of global misery, but also limit our potential as a species to collectively solve our problems—in a sense, humanity is currently in the position of trying to save itself with one hand tied behind its back. Similarly, global injustice is not only a direct driver of hazards like climate change and global conflict, it also increases civilizational vulnerabilities and exposures to these hazards by disempowering those at the risk frontlines, creating structural inertia that hampers risk mitigation and by slowing recovery from disasters [Liu et al., 2018; Avin et al., 2021]. Moreover, how we deal with non-existential but still-catastrophic threats is likely to provide lessons (or warnings) about how we will deal with existential ones, the Covid-19 pandemic being an example [Liu et al., 2020]. As a consequence, even conventional ‘long-termist’ ERS accounts³⁸ have already recognized strong instrumental arguments in favour of reducing present-day inequality, on the basis of flowthrough effects on existential risk reduction in the future [Schmidt and Juijn 2021].

Furthermore, the debate over whether to prioritize existing problems or rather focus on future ETRs raises a further question: whether existential risks should be seen as relatively exogenous (something out there that could strike us down, like the proverbial asteroid), or as an intrinsic feature of wider problems in our present society and culture, that need to be addressed today as a precondition of being able to mitigate the specific ETR hazards at some future point. For

³⁷ See for instance Ord et al. [2021] and Maas et al. [2021].

³⁸ For more discussion of these approaches, see Question 3.

example, the risk posed by the development of Artificial General Intelligence (AGI) at some point in the future has been argued to be amongst the biggest technological threats to humanity, larger than threats like the economic concentration of technological power and the risk of unemployment due to automation right now.³⁹

However, while smaller in impact, these intermediate risks are potentially important sites for ERS scholars to engage, since debates over who develops this technology, to what end, and with what safeguards, may prove just as salient to the technology's long-term societal impacts and risks as the more technical considerations of safely aligning such a system with human values. In fact, there are good reason to believe that AGI risks will be much harder to address, technically and especially politically, if earlier action is not taken to address existing or near- and medium-term risks around AIs, or if we fail to find practicable and scalable solutions to issues of societal value alignment [Gabriel, 2020; Gabriel and Ghazavi, 2021].

When it comes to technological development specifically, aligning ERS with research focused on existing challenges might become a little more complicated, and there is a greater possibility of tension. Seth Baum [2015] describes two ways of such alignment, through exploiting co-benefits or through mainstreaming. However, many of the disciplines or stakeholders where we might turn to to find solutions to emerging technological risks, might ironically also constitute some of the sources or contributors to these risks.

³⁹ Although it should be noted that in recent years, ERS scholars have taken a more plural views on the risk scenarios around advanced AI [Clarke et al., 2021], and some have focused at length on challenges from power concentration and catastrophic outcomes that would not necessarily involve AGI- or superintelligence-like systems [Bostrom et al., 2019; Critch and Krueger, 2020]. More broadly, the distinctions between 'near-term' and 'long-term' AI concerns may be overstated or oversimplified [Prunkl and Whittlestone, 2020; Stix and Maas, 2021].

Addressing potential risks of ERS research itself?

Of course, there is also a distinct question to be asked, which is: are there ways by which ERS research is not so much ‘useless’ (in terms of option value), but where it could rather contribute to ETR-scale harm? Interestingly, ERS research, or mishandled high-profile ETR mitigation programs, could be harmful in various ways.

For one, in some cases, technical research relevant to risk mitigation might directly contribute to the risk, as has been suggested for AI research that might prove of key importance in learning how to safely align this technology—but which simultaneously helps contribute to capability progress, therefore potentially speeding up the very arrival of the risks, and reducing the amount of time available for others to undertake such research [Krueger, 2019].

Alternatively, ERS research could create ‘information hazards’ [Bostrom, 2011], or framing risks. This could happen if ERS research, or the way it is framed, inadvertently focuses the attention of certain actors (e.g. states) on the prospect of certain risky-but-strategically-tempting technologies (e.g. advanced AI), in ways that could perversely lead to self-fulfilling prophecies or arms races [Cave and ÓhÉigeartaigh, 2018]. Furthermore, it has been suggested that identifying some particular ETRs or framing them as threats to ‘existential security’⁴⁰ may prove counterproductive, since this need not drive actors towards more effective responses (e.g. global coordination), and may instead drive policies that are not necessarily tailored to- or even all that relevant to the mitigation of the underlying risks, while potentially coming at large normative, political or democratic costs [Cremer and Kemp, 2021: 22-23]. More generally, while it may be necessary to undertake conceptual and strategic discussions over potential emergency interventions to respond to certain imminent ETRs, proposals for decisive yet far-reaching and irreversible ‘pivotal acts’ may also

⁴⁰ [Sears, 2020]; see also the discussion of the emerging ‘existential security’ approach to ERS, under Question 3.

introduce novel sources of risk or instability if they are not well judged, or if they fail in non-graceful ways that compromise future or parallel routes to ETR mitigation.

Thirdly, even counter-ETR mitigation programs that are in principle effective might inadvertently induce risk homeostasis or raise a kind of *moral hazard*.⁴¹ This relates to the idea that working to make ourselves safe in conspicuous ways, can perversely make us less so. The idea of ‘moral hazard’ has recently become associated with an objection to the development of geoengineering as a potential response to climate change: in this context, some critics fear that even modest research in the area might come to represent an illusory safety net about climate change, and encourage complacency around the steps that need to be taken now to mitigate this risk [Wagner and Merk, 2019]. Others have suggested that moral hazard concerns around geoengineering are important but defeasible [Currie 2018]. On the other hand, Dasgupta has made the broader claim that even to focus specifically on the risks of anthropogenic climate change may lead to a form of moral hazard, by focusing on too narrow a problem, and lead to solutions that will not be sufficient to protect the earth’s biosphere as a whole [Dasgupta, 2019].

Similarly, approaches to managing ETR that are too narrowly focused on specific threats may risk giving a false sense of security if broader systemic vulnerabilities and exposures to existential of global catastrophes remain unresolved, and it is important that our desire to encourage investment in risk management solutions that we are aware of (or even hopeful about) does not lead to an unrealistic representation of how safe these might make us.

⁴¹ The term ‘moral hazard’ is taken from insurance, where it is seen that people who insure themselves against certain risks may take fewer precautions to prevent them. It however has roots in far older, medieval theological and probability literatures [Rowell & Connelly, 2012].

Question 3) What are the different approaches to studying and managing ETRs?

Notwithstanding the many ways in which researchers seek to align their work on ETR with one another, ERS remains diverse in its methods, approaches, and assumptions. In this section, we will aim to both distinguish a number of distinct epistemic communities or scientific traditions in the ERS space, before finally briefly discussing the academic and institutional ‘substrate’ (e.g. the organizations and universities) within which the ERS community operates.

Unsurprisingly, approaches to ERS can vary hugely; yet, in practice, it is possible and useful to consider how many distinct approaches can be distinguished by their diverging positions along two axes, relating to a researcher or community’s views on (1) *the role of ethics* in our understanding of ETRs, and (2) the most appropriate *theory of change* for managing such risks.

On the former question around the role of ethics, we distinguish between paradigms that (implicitly or explicitly) see the field of ERS as tied up with particular ethical worldviews (especially those which put a particular premium on ETR mitigation), versus those that aim instead to decouple the study of ETRs from any particular ethical program.⁴²

⁴² Cremer and Kemp [2021:12], for instance, have argued that ERS should more clearly separate ‘the science of risk from the moral evaluation of risk’ [2021:18]. Specifically, they draw a distinction between the study of (1) ‘extinction ethics’, which explores the ethical implications of extinction; (2) ‘existential ethics’, which explores the ethical implications of different societal forms and futures in order to understand what (non-extinction) events or trajectories should count as ‘existential risks’ (e.g. dystopias under most value theories); and argue that the ethical studies of these futures should be separated from (3) the ‘analysis of human extinction and global catastrophe’ per se [2021:12].

Interestingly, while they critique what they see as the unreflexive intertwining of (long-termist) ethics with analysis of risk in ERS, their argument also presents a distinct and preferred approach to analyzing and mitigating

On the latter question of the most appropriate ‘theory of change’, we distinguish between ‘*inside working*’ (working as a trusted partner with established institutions or frameworks or actors, including those who may be directly or indirectly implicated in the creation of risks, as with militaries operating nuclear arsenals) and ‘*outside working*’ (making independent statements about the nature of risk without concern about how these might be received). While questions of the role of ethics, and the theory of change are distinct questions, which communities can debate separately, within the ERS community they are often intertwined and so it is relevant to discuss them together in this way.

Surveying the intellectual landscape

ERS is a field notable for both the independence of its researchers and their interest in cross-disciplinary collaboration. Thus there really are no categorical distinctions. However, from a genealogical perspective it is possible to differentiate between at least four loosely related perspectives to the study and management of ETRs that take different positions on these two questions. These perspectives are (1) long-termist; (2) systems-theory; (3) existential security; (4) concerned science. As with the community as a whole, it is noteworthy that many

ETRs, one which itself appears deeply grounded in theories of deliberative democracy [Cremer and Kemp, 2021:25-27]. The resulting ‘democratic approach to ERS’, to be sure, is not presented as an ethical framework in its own: rather, it is instead advocated (1) on what appears to be meta-ethical grounds (i.e. as an approach that provides greater room for different ethical and empirical assumptions), (2) on analytical grounds (as offering important and neglected insight about the frequently state-based origins of many ETRs); and (3) on grounds of the superior instrumental benefits of democratic approaches (e.g. societies and institutions) in mitigating ETRs. These may be all compelling arguments; yet, without trying to adjudicate them directly, it remains an open question whether- or on what grounds this democratic approach to ERS successfully or meaningfully avoids introducing its own ethical precommitments into either the analysis of the most important or culpable sources of ETRs, or of avenues towards their mitigation.

whose work is aligned with ERS as a discipline may not align their work specifically with any of these paradigms, or may be aligned with more than one, while work may be of significant interest to researchers who focus on a particular paradigm even if it does not originate from within that paradigm. Still, we feel that understanding these different perspectives is helpful in assessing the epistemic diversity of the community of ERS.

The first, ‘long-termist’ school of ERS has been a prevalent and particularly influential view within the study of ETRs.⁴³ Indeed, it may well be the ‘original’ approach of the modern field of ERS as it has developed in the last few decades. It has been primarily developed through the work of Nick Bostrom, Eliezer Yudkowsky, and other early contributors in the Rationalist (and before that ‘extropian’) communities [Chivers, 2019; Moynihan, 2020], including early work that first coined

⁴³ Cremer and Kemp [2021:2] have alternately described this as the ‘Techno-Utopian Approach’ (TUA), which in their analysis is based on ‘three main pillars of beliefs: transhumanism, total utilitarianism and strong longtermism’. On the other hand Beard and Torres (2020) differentiate between two distinct ‘waves’ within this paradigm, with the first wave being marked by strong commitments to transhumanism and the second by the dropping of these commitments together with the significant expansion of the field precipitated by the growing support of the Effective Altruism community. In this chapter, however, we refer to ‘long-termist’ as the more specific, arguably most crucial, and most long standing philosophical component of this approach to studying ETR. Even so, it should be noted that the long-termist approach to ERS is diverse, and finds itself at the intersection of a complex overlapping set of philosophies and communities that include not just those focused on specific existential risks directly, but also the broader Effective Altruism movement, the Rationalist community, and, most recently (if only partially), the emerging field of Progress Studies. Importantly, these communities are not fully coterminous, and often have important disagreements [Crawford, 2021; Daniel, 2021]. Given their overlap and engagements, we do discuss them together here under the ‘long-termist’ label, however, some of the points we make in this chapter apply more strongly or directly to one or the other communities, which we flag where appropriate.

terminology around existential risks, and linked these closely to technological development [Bostrom, 2002].

The long-termist view sees ERS as a field that is closely aligned to and entwined with particular ethical values or axiologies. That is, much of this work is motivated by- and advances a ‘strong long-termist’⁴⁴ form of consequentialism (or ‘time-neutral’ total aggregate utilitarianism) as basic ethical theory. This ethical view is today perhaps the most developed and internally coherent rationale for studying and mitigating ETRs today.⁴⁵ This pre-eminence may in part be because the emphasis of total aggregate utilitarian normative theories on inter-temporal neutrality in effects can naturally underwrite a concern with mitigating existential risks [Baum et al., 2019]—although, as these scholars themselves also note, an utilitarian ethical orientation, or a

⁴⁴ It should be noted that while Bostrom [2003] already advanced an ‘aggregative person-affecting’ account of the challenges of human extinction, and Beckstead [2013] made an influential argument that positively influencing the far future has high moral importance, the term ‘long-termism’ itself is relatively new, dating from 2019 [MacAskill, 2019] to encapsulate a cluster of views that hold that “the most important determinant of the value of our actions today is how those actions affect the very long-run future.” [MacAskill, 2019]. This has been deepened into a comprehensive theory of ‘strong long-termism’, with an authoritative statement being Greaves and MacAskill’s [2019].

⁴⁵ To be sure, there remains considerable ongoing debate and development around the topic of longtermism. This has included exploration of various epistemic challenges [Tarsney 2019]; critiques [Masrani 2020; Cremer and Kemp 2021] and responses or rebuttals [Aird 2021b; Balwit 2021], discussions of common misconceptions about long-termism [Malde, 2021], debates over different theories of change or impact around “working toward a better world” in the long-term [Karnofsky, 2021b], as well as recent further developments or extensions of the theory, such as calls for ‘regular long-termism’ [Manheim, 2021a], or (historicist-focused) ‘Burkean longtermism’ [Aschenbrenner 2021], to name a few.

concern with the far future, are not strictly necessary to justify concerns about ETRs [ibid.; Baum, 2015; Wiblin & Harris, 2021].⁴⁶

As such, while an important school of thought, the ethical assumptions in the long-termist school of ETRs studies are not exhaustive of all ethical perspectives on existential risks; indeed, they are also likely (for better or worse) not representative of global public ethical views on these questions [Schubert et al., 2019; Cremer and Kemp, 2021]—nor, for that matter, of moral philosophers. For instance, surveys of the Effective Altruism (EA) community (which is not entirely coterminous with the ERS community, but which has advanced significant concern for existential risks) suggest that its members put greater stock on consequentialist theories of ethics, more than do moral philosophers [Cremer, 2020].⁴⁷ While scholars in the long-termist ERS community have, to their credit, emphasized the importance of ‘epistemic modesty’ [Lewis, 2017] as well as the importance of exploring new assumptions and perspectives, given our background condition of ‘moral uncertainty’ [MacAskill et al., 2020], the paradigm’s idiosyncratic meta-ethical commitments mean that there is a risk that these long-termist approaches to ERS might become homogeneous and insular if they do not sufficiently engage with other views.⁴⁸

⁴⁶ For instance, see [Beard and Kaczmarek 2022], elsewhere in this volume, for one account of how similar views can be defended from the perspective of a very different ethical tradition, social contract theory.

⁴⁷ By comparing PhilPaper Surveys against EA community surveys, Cremer reports that “25% of philosophers lean towards deontology, 18% to virtue ethics, 23% to consequentialism, [...] EA’s give only 3% (2015, 2019), 2% (2014), 4% (2017) to deontology, 5% (2015, 2014, 2017), 7% (2019) to virtue ethics and 69% (2015 and 2014), 64% (2017), 81% (2019) to consequentialism.” At the same time, it is suggestive that these surveys also show that it is not the case that any other theory of ethics commands a philosophical supermajority.

⁴⁸ But for exceptions, see also Baum [2015], as well as Baum and Handoh [2014] on integrating ERS with the Planetary Boundaries framework; or Sandberg [manuscript] which surveys a landscape of potential ‘grand futures’, and in doing so engages with a range of meta-ethical orientations, including

Moreover, in the long-termist approach, both the threat model of ETRs, as well as the ethical grounds for why these risks should be mitigated, are closely connected to the anticipated risks and promises of emerging technologies. In particular, technologies such as AI, robotics, and nanotechnology are seen as centrally important to the future of humanity, for both good and bad [Karnofsky, 2021a].

On the one hand, this approach generally holds that technological growth and innovation are at the very least a net good for global welfare and wellbeing in the near-term,⁴⁹ and because continued technological development is assumed to eventually deliver the technologies (or the societal wealth) necessary to address or mitigate ETRs;⁵⁰ but also because this approach specifically predicts that technologies such as AI will be so transformative that they are the best way to eventually achieve truly long-lasting survival of humanity [Baum et al., 2019]; a technologically ‘mature’ civilisation [Bostrom, 2013: 19]—or even a post-human future—that could maximise our future potential, and realize “astronomical” amounts of future value [Bostrom, 2003].

At the same time, and perhaps accordingly with its view of the high technological stakes, the long-termist ERS approach also recognizes that many of these selfsame technologies which could bring plenty may also pose terminal risks to humanity, especially as we remain in a current ‘semi-anarchic default condition’ [Bostrom, 2019:457] situation. For the long-termist ERS school, this grounds the rationale that new approaches or interventions to managing these risks are required [Rees, 2003; Bostrom, 2013; Ord, 2020]. In doing so, the long-termist approach may take a view of the technological sources of ETRs as relatively

those that emphasize intergenerational justice and sustainability, humble futures, and many others.

⁴⁹ That is, “a default presumption that economic and technological development is positive” [Beckstead, 2015].

⁵⁰ See for example the argument made by Aschenbrenner [2019].

exogenous. It considers the development and adoption of many risky technologies as something that may ultimately be hard or even impossible to hold back or relinquish for good, if that were even desirable. In taking this view, the long-termist approach has been critiqued as ‘techno-determinist’ [Cremer and Kemp 2021], though it might be more specifically understood as a worldview that assumes ‘military-economic adaptationism’ [Dafoe, 2015]:⁵¹ this theory, a special case of sociotechnical selectionism, assumes that while technologies do not have intrinsic and linear developmental paths, and may be shaped or restrained by social choices in local contexts in the short run, in an anarchic world, the pressures of economic and military competition will gradually and over longer time-frames constrain sociotechnical evolution to relatively deterministic paths.⁵²

As such, the long-termist approach to ETRs on the one hand still focuses a lot on attempts to ‘steer’ the direction and pace of technological progress [Karnofsky, 2021b]—in particular paying attention to ways in which defensive or protective technological developments might be sped up, and offensive or dangerous technologies slowed down, in order to avoid risky combinations.⁵³ Yet on the other hand, the

⁵¹ As also discussed in Cremer and Kemp [2021:17]. However, for a genealogy of the concept of ‘technological determinism’, and a critique of the discursive ways that that term is used in academic debates over technology, see also Peters [2017].

⁵² As Dafoe [2015:1059] defines it; “Military–economic adaptationism states that (C1) when there is sufficiently intense and prolonged economic and/or military competition, and given that (C2) technology enables new sociotechnical configurations, (C3) some of which confer advantages in economic/military competition, sociotechnical systems will evolve to become more adapted to success in the economic and/or military competition.”

⁵³ An approach that has been called ‘differential technological development’ [Bostrom, 2002; Beckstead, 2015]. For an expanded taxonomy which relates this concept to the larger categories ‘differential intellectual progress’ and ‘differential progress’, see Aird [2020]. Cremer and Kemp [2021: 17-18] have critiqued the concept of differential technological development, arguing that it

long-termist approach also remains aversive to or pessimistic about the prospects of unilaterally halting underlying scientific developments, even those that contribute to ETRs [Balwit, 2021]. Tellingly, this is not because they consider it categorically impossible for humanity to permanently curtail some technologies that give rise to ETRs.⁵⁴ Instead it is assumed that military-economic adaptation would gradually converge towards many ETR-related technologies (for their strategic merits), and that successfully containing such competition (as well as monitoring for rogue programs) would require an incredibly strong level of global coordination and control, or new technologies, beyond that achievable today.⁵⁵

In making such analysis, it is possible that this approach is too pessimistic over the prospects of curtailing or limiting the proliferation of new technologies, even those considered of immense strategic use.⁵⁶ At the same time, global governance scholars have long recognized the difficulties of assuring global public goods such as ‘mutual restraint’ [Barrett, 2007; Van Aaken, 2016]. As such, particularly in a time of fraying international cooperation, the underlying question of whether or how we might ever ensure sufficient global coordination to guarantee

has remained relatively undertheorized and under-justified, and that it may not offer grounds for interventions compared to straightforward moratoria or bans.

⁵⁴ Though such proposals for bans do often give rise to separate concerns over the lost future value of the future—or value lock-in.

⁵⁵ This includes views that it might require forms of world government [Bostrom 2019]; or one actor utilizing advanced AI technologies to establish a ‘singleton’—‘a world order in which there is at the global level a single decision-making agency’ [Bostrom, 2014b:78]. Cremer and Kemp [2021] have critiqued the long-termist (or what they call the ‘Techno-Utopian’) approach on these grounds, arguing that the centralization of power, and the affording of sweeping surveillance systems, would increase rather than reduce ETRs.

⁵⁶ See for instance Maas [2019], discussing the historical track record of nuclear non-proliferation and arms control, which is imperfect but surprisingly robust (especially when compared against *ex ante* expectations in the 1960’s perspective).

sufficient restraint over sufficiently long timeframes (or potentially indefinitely, into an extremely long future), to indefinitely forestall strategic technology-generated ETRs, remains an important challenge for critics of the long-termist approach to answer.

In contrast to its strong links in terms of ethics, the long-termist ERS approach has not yet articulated a conclusively determined theory of change. Broadly speaking, it is relatively neutral between ‘inside working’ as a trusted partner with established institutions and ‘outside working’⁵⁷ where most of its researchers focus on purely academic research, but may have a strong desire to influence groups who are open to collaboration, such as major governments, tech companies or other key stakeholders.

A second approach that has emerged more recently tries to decouple the study of ETR from any particular ethical program, studying phenomena like human extinction and global catastrophic risk from an analytical point of view, and leaving the normative implications or recommendations to separate analysis.⁵⁸

Researchers in this paradigm tend to be interested in technological risk from a *systems theory approach* [Avin et al., 2018; Kreienkamp and Pegram, 2020], perceiving risks as emerging from the interaction between technological developments and critical systems required for human survival and flourishing [Liu et al., 2018; Manheim

⁵⁷ Although it should be noted that many in this community have on occasion been concerned about the potential impacts of making certain risk framings—as seen in the field’s concern over the spreading of ‘information hazards’ [Bostrom, 2011], or the risk that certain descriptions of technological risks could contribute to adverse ‘arms race’ framings that actually increase both ETRs as well as other harmful effects [Cave and ÓhÉigearthaigh, 2018].

⁵⁸ See also notably Cremer & Kemp [2021]. Beard and Torres [2020] combine their discussion of this and the following approach to studying existential risk into a pluralistic ‘third wave’ of the field’s development, although they are both ethically and epistemically distinct.

2020]. Some who share this approach⁵⁹ are also committed to a theory of change based on outside working, which they claim helps to preserve academic integrity and independence from the influence of third parties with a vested interest in these systems, while taking a critical stance towards those they see as primarily responsible for producing such risks. These researchers may point to the extensive history of corporate action aimed at downplaying or obfuscating risks, and the resulting important role of independent research in overcoming this (e.g. Oreskes and Conway [2011]). Others however take a more neutral or pragmatic view to combining different theories of change, or see inside working as important to actively correct for other biases (such as ideological commitments or an oversimplified theory of institutional behaviour).

A third approach to ERS sees ETR as primarily a matter of *security*. This paradigm draws close connections between ERS and longstanding work in already securitized fields such as biosecurity, cybersecurity, sustainable security, or nuclear security. Although there are many different approaches to the securitization of existential risk, one particularly prominent recent example has been the development of the concept of “existential security” [Sears 2020]. This combines a concern about extreme technological risk with existing security paradigms with a focus on removing specific vulnerabilities and threats. It may involve scholars aiming to work within and with national security actors. The benefit of a securitization viewpoint is a greater leeway to engage with existing security-based institutions in governments and militaries; these actors often already work on many aspects related to extreme technological risk, they may be the critical or even the only gatekeepers in achieving any mitigation in some risk areas, or they may at least be able to bring to bear unusual resources, expertise, or political attention.

⁵⁹ This is particularly the case for scholars working in the adjacent but overlapping approach of ‘critical studies’. However, while an important and emergent approach to ERS, with distinct views, the ‘critical ERS studies’ approach is still young, and it requires further work and discussion.

In some politicized issue areas, the involvement of national security actors could even provide political capital or credibility that stretches across prevailing party lines or political coalitions [Klare and Perry, 2021].⁶⁰ However, in engaging with military institutions, the paradigm opens itself up to many pre-existing points of criticism about such actors, such as their tendency to be overly threat focused, their dependence on the military industrial complex, or the risk that security-frame solutions might impart zero-sum thinking and unilateral national policies that undercut global cooperation on key risks.

While the above paradigms were developed or adapted specifically for the study and management of ETR as a whole, the oldest, and perhaps best known, paradigm in ERS is importantly distinct: that of the *'concerned scientist'*. Ever since scientists involved in the Manhattan Project initiated projects like the Bulletin of the Atomic Scientists,⁶¹ there has been a strong tradition of concerned scientists and researchers seeking to prevent technological catastrophes by democratising knowledge about the risks inherent in their own field and calling on politicians to act to promote more (technological) safety. Often, members in this perspective do not start with a concern about existential risk per-se, but rather have developed a deep understanding of some scientific issue or domain, which leads to them being concerned about a threat related to that issue in particular; only further along may they potentially find common cause with others in ERS. As a result, work in the 'concerned scientist' paradigm is even more diverse than those already discussed, yet often shares a deep commitment to pluralism and democracy and a theory of change that focuses on moving from inside work (if this is perceived as being ineffective) to outside work.

⁶⁰ Even in ETR domains where militaries are not yet active, it may be important to engage with such actors in advance, to ensure that sudden developments in those other domains (e.g. algorithmic flash crashes) will not trigger poorly tailored, sudden escalated reactions from them.

⁶¹ See also [Beard and Bronson, forthcoming], which deals with the Bulletin of the Atomic Scientists' track record.

Historically much of this work focused on the risks of nuclear war, though there have always been those concerned about other technologies as well, from the environmental impacts of industrial farming [Carson, 1962] to the potential for runaway AI [Good, 1964] or nanotechnology [Drexler, 1986] to some day threaten humanity.

Surveying the institutional landscape

Having discussed four distinct epistemic communities in ERS, we now turn briefly to discussing the underlying institutional landscape and academic infrastructure across which these communities interact. Notwithstanding this ethical and epistemological diversity of approaches to ERS it is also worth noting that there is much within the intellectual heritage and institutional placement of this community that both binds researchers together, but can also make it harder for new researchers or perspectives to enter the field. For instance, a disproportionate share of attention within ERS still seems to be paid to the work of a small number of research institutes associated (directly or indirectly) with leading global universities, such as the Future of Humanity Institute (FHI, Oxford), Centre for Human Compatible AI (CHAI, UC Berkeley) Center for Health Security (CHS, Johns Hopkins) or the Centre for the Study of Existential Risk (CSER, University of Cambridge). Similarly, early entrants into the field came disproportionately from a small number of disciplines, such as philosophy, economics, and engineering. This not only affects who is undertaking this research⁶² but also how and where they were trained, what kind of intellectual assumptions they are likely to share, and which network are more or less connected with the community.

This institutional concentration might exist for several reasons. The straightforward explanation could be that these institutes produce a disproportionate share of original research in the field, simply because

⁶² A question we will take up in Question 6, below.

they are often larger and better funded than other organizations, are longer established and have had more time to operate, are able to attract the leading researchers to work for them—and/or may foster, train and aim to attract the community of researchers who actively identify with ERS. Moreover, given how young the field is, it is possible that there is a founder effect, such that concentration is not strange.

Alternatively, the institutional spread of ERS may be the product of peculiar scientific incentives: it may be that the public credibility of elite institutions is better able to afford academics with the freedom and leeway to explore more tentative topics—including culturally charged, dramatic topics like human extinction—that might be perceived as ‘unserious’, too ‘large’, ‘speculative’, or too risky to bet an academic career on in other, less well-resourced institutions or more precarious disciplines. In this way, rather than it be the case that only academics at elite institutions have been fascinated by these questions, it may instead be the case that elite institutions simply provide more shielding against a range of academic incentives or properties which otherwise, and at many smaller institutions, strengthen the trend towards ‘scientific conservatism’ [Currie, 2019], and throw up institutional barriers to those who have sought to study such topics there.

More critically, this pattern could also reflect the fact that these are elite and privileged institutions, which benefit from their position in unjust and unjustifiable global hierarchies of wealth, race, and status. That need not imply that these institutions—or their work—are not deserving of the attention that they receive, especially given the importance of the subject that they study. However, it does reflect the fact that they are often able to produce more and better research precisely *because* of their privileged access to resources. We should be glad that there are organizations occupying these privileged positions who use their status and authority to study risks facing all of humanity but that does not mean that we would not wish to design things differently if we were able to from scratch. This is particularly the case because important

and relevant work is often done in other disciplines and universities, and the field should gain immensely from marshaling that talent.

In sum, we see that, while there is already quite a high degree of diversity amongst ERS scholars about many questions concerning the study and management of ETR, that diversity could likely still be substantially increased were a greater plurality of voices, perspectives and institutions to enter the field. Indeed, while it may often appear appealing for an ERS community to work with those that share or agree on a series of fundamental assumptions or reference points, this risks breeding implicit assumptions or blind spots:⁶³ as noticed by Cremer [2020], counteracting trends towards intellectual homogeneity is critical to ensure corrigibility mechanisms to communities focused on ETR mitigation, such as the Effective Altruism community, because “intellectual homogeneity is efficient in the short-term, but counter-productive in the long-run. Value-alignment allows for short-term efficiency, but the true goal of [Effective Altruism] – to be effective in producing value in the long- term – might not be met.”

We believe that, as it stands, there is no one best approach to ERS and that the traditions of both independence and collaboration that have emerged in the field are worth continuing, even if their different theories of change can sometimes create tensions between researchers and their intermediate objectives. Ultimately, the four approaches we have sketched—the (1) long-termist; (2) systems-theory; (3) existential security; (4) concerned science—might once have been somewhat foundational, but should not be considered archetypical. As disciplines involved have expanded, and the number of people involved has increased, we should expect individual researchers to draw from these (and other) traditions, along with bringing in new approaches to the area. In general, a likely direction of travel for ERS will be the continued

⁶³ Though for reflexive critical discussions from within the ERS community itself, exploring potential blind spots, see also Ellen [2021] and Critch [2021].

increase in approaches and underlying motivations or ethical frameworks. In doing so, the field should critically consider the question of what this will mean: how (or even whether) to aim at ensuring a level of coherence and consistency to what the field is—and whether it can, or should, be sustained as a ‘field’.

Question 4) In what ways can the field of ERS pursue a coherent scientific approach to ETRs?

Historically, ideas around ETRs have been influenced by many disciplines, including the humanities and a long tradition of science fiction writing about such subjects. As well as the diverse methods and approaches brought from individual disciplines, we now turn to considering what approaches ERS could use collectively within a more scientific approach to understanding ETRs

Indeed most people within ERS today, whatever perspective they take, would aspire to study these risks scientifically. Such claims can be understood in at least three senses: (1) scientific as opposed to humanistic (i.e. employing the scientific method); (2) scientific as opposed to speculative (i.e. making claims to be non-fictional); and (3) scientific as opposed to pseudo-scientific (i.e. being truth tracking and falsifiable in its claims). It is natural for the community to want to advance all three of these claims about its work; however, the confluence of unprecedented new technologies and irreversible global risks could make this hard to sustain. At present, three methodological developments are commonly cited by different parts of the ERS community as providing a basis for claims to be scientific: (1) the robust use of evidence, particularly through falsifiable predictions; (2)

engineering and risk-assessment tools; and (3) the use of futures and foresight techniques.⁶⁴

The standard model of scientific inquiry holds that scholars and scientists in any field ought to restrict themselves to *making hypotheses that can then be subject to experimental falsification* [Popper, 2008]. This is often seen as the very epitome of the distinction between science and pseudo-science. However there might be a clear barrier to adopting this approach with regards to ETRs—because at its core, this involves a series of hypothesized scenarios that *we do not want tested*. As Carl Sagan once put it in relation to one particular existential risk: “[u]nderstanding the long-term consequences of nuclear war is not a problem amenable to experimental verification – at least not more than once” [Sagan, 1983]. That is not to say of course that all aspects of studying ETRs are entirely impervious to this approach, of course. For instance, in establishing characteristics for effective governance arrangements or resilient systems, and in comparing existing systems to these, it is possible to disprove hypotheses about the workings of those idealized systems.

More broadly, it is of course possible for ERS scholars to make predictions about issues that are related or adjacent to ETRs, or that would alter their subjective probability estimate for an ETR, but would not require humanity going extinct in order to adequately test them. Still, as scientists who have attempted to formulate such predictions have found, this does face further epistemic problems, namely (1) the challenge of appropriately operationalizing meaningful proxies for catastrophe predictions, and (2) the challenge of interpreting their resolution. For instance, as a proxy for demographic catastrophe stemming from population growth and resource insufficiency, Paul Ehrlich famously bet the economist Julian Simon that the value of

⁶⁴ See also Sundaram and Currie [2022, in this volume]; as well as [Kemp, 2022, in this volume].

copper, chromium, nickel, tin, and tungsten would rise between September 1980 and September 1990, while in 2002, Rees proposed a bet that “A bioterror or bioerror will lead to one million casualties in a single event within a six month period starting no later than Dec 31 2020”—a wager which was accepted by Stephen Pinker [Rees and Pinker, 2017]. While Erlich lost his bet and Rees and Pinker agreed not to settle due to uncertainty about the origin or the COVID-19 pandemic [Rees, 2021], in both cases there is an argument to be made that a failure to win does not invalidate the proposers’ broader concerns,⁶⁵ because the points they were seeking to draw attention to remain valid and important regardless of the outcome; the goal of their predictions was neither to prove or disprove a particular theory, but rather to raise attention to potential sources of catastrophic risks (resource depletion and biorisks) that are highly uncertain in their particulars, but relatively certain or plausible as general processes of risks that have independent sources of evidence or support; both scientists succeeded in making their point.

Against such arguments however should be set the epistemic value of making and resolving well-operationalized predictions, and fairly updating our beliefs on their outcomes, if only to keep us from (a) selectively latching onto our predictions and bets only when they are right, while (b) always moving the goalposts when they do not quite come out right, on the rationalization that we were somehow ‘right in spirit’. Indeed, parts of the ERS community have put strong emphasis on the epistemic value of well-adjudicated predictive claims in order to clarify uncertainty and adjudicate disagreements.⁶⁶ Different approaches

⁶⁵ Though, from a Bayesian perspective popular within some parts of the ERS community, it might still count as—weak—evidence against their particular claims.

⁶⁶ Indeed, the practice of turning disagreements or debates into concrete predictions, and even staked bets, has become a feature and practice in parts of the ERS-overlapping Rationalist and Effective Altruism communities [Yudkowsky, 2007]. In particular, it has been argued that their affinity for

to when or where (quantitative) predictions are warranted, and how to treat their outcomes, represents a potential point of epistemic tension between different researchers in this space.

A second approach to scientifically studying extreme technological risk can be taken from *engineering and risk assessment tools*, such as ‘fault trees’. These seek to assess risks that are unlikely to occur but would be catastrophic if they did by first identifying fail states in the system being studied and then working backwards to identify what parameters would need to hold in order for such a fail state to be achieved. While not necessarily applying the scientific method as such, these approaches aim for a degree of analytical rigor and generalizability that allows them to present an objective assessment of risk beyond that typically discussed in the social sciences and humanities. Still, this method is far from bulletproof: engineers might incorrectly identify failure states, miss important failure states, incorrectly determine what conditions would cause a failure state to occur or miss failures that can be caused by the operation of a system as a whole (which is harder to model) rather than the operation of its specific parts. In some cases, the susceptibility of complex, tightly coupled sociotechnical systems to so-called “normal accidents” [Perrow, 1982] may mean that the straightforward addition of fail-safes may perversely increase overall risk, by increasing the total complexity of the system, increasing the number of failure or attack points, or increasing operator overtrust and automation bias [Maas, 2018; Williams and Yampolskiy, 2021].

Nevertheless, fault-tree risk assessments can be valuable for risk management processes because they provide a systematic way of thinking about risks, how they occur, and how they can be prevented. Specifically, they allow researchers to pinpoint useful intervention points (e.g. at earlier stages or at lower cost); gaps in existing interventions (by

probabilistic and predictive reasoning under uncertainty supported these communities’ early reaction to the COVID-19 pandemic [Alexander, 2020].

mapping intervention options to fault trees), or gaps in knowledge or previously under-appreciated areas of uncertainty (e.g. through expert elicitation). Fault Trees can also be augmented by the addition of conditional probabilities for each step in the tree given what other steps have occurred, in which case they are known as Bayesian Networks. Finally, fault-trees present a view of these risks in ways that are easy to interpret and thus build on or respond to. Some especially well-developed fault trees for thinking about ETR include Barrett and Baum [2016] on AI risk, and Baum, de Neufville and Barrett [2018] on nuclear war risk.

A third approach to the study of ETR that carries different scientific credentials is the use of *futures*, *foresight*, and *horizon scanning*. While these terms can be used to describe a wide variety of techniques, the fundamental aim in each case is to provide a systematic way of thinking about future events that allows the judgments of multiple people to be compared and combined. These approaches aim to move beyond individual speculation about what might happen to describe a possibility space or consensus judgment that draws together all of the available information while filtering out individual biases and other sources of ‘noise’. For instance, the Centre for the Study of Existential Risk has adapted techniques developed in conservation horizon scans, for studying ETS [Rios Rojas et al., 2021]. This leads a group of risk assessors through a structured expert elicitation approach. Here, they are first invited to share their prior beliefs about a particular ‘future-focused theme’ by submitting for the group’s consideration a range of specific topics they believe to be important. After an initial round of scoring, the experts then discuss the topics and learn from one another before then independently re-evaluating all the submissions again based on this new information. The overall aim is of converging towards a shared opinion-set. Anonymous information about all of these steps can then be presented in a way that makes the process transparent and replicable. The modified version of this technique used by CSER and others is based on

the IDEA (Investigate, Discuss, Estimate, Aggregate) protocol [Hanea et al, 2017], itself drawing on a long heritage of similar techniques dating back to the Delphi, or ETE (Estimate, Talk, Estimate), Method developed by RAND in the 1950's and 60's. Anonymous information about all of these steps can then be presented in a way that makes the process transparent and replicable. CSER has used this technique to explore emerging issues in biological engineering [Wintle et al., 2017; Kemp et al., 2020], and in UK biosecurity [Kemp et al., 2021].

A final popular technique for understanding extreme technological risk is Superforecasting, developed by the Good Judgment Project [Tetlock and Gardner, 2016]. In a further development of these techniques, CSER has also begun to develop interactive scenario-based tools and role-play exercises which serve both to help understand different stakeholders' understanding of and relationship to catastrophic risks such as from emerging technologies, in order to encourage participatory technology assessment engagement from a wider group of experts as well as citizens [Avin et al., 2020, Cremer & Whittlestone 2021].

Question 5) How can or should ERS scholars achieve influence or impact to mitigate ETRs?

Throughout academia, the notion of achieving 'impact' has become increasingly prominent. This is particularly the case for ERS scholars. The tagline of CSER is that we are involved in "the study *and mitigation*" [emphasis ours] of existential risk". 'Mitigation' is an action, an intervention, that by its nature goes beyond research. Participating in that action is part of the job description yet, as discussed, there are multiple competing theories of change amongst those who seek to study extreme technological risks, which can give very different prescriptions for how to achieve this mitigation.

A first approach might be roughly termed the ‘elite spokesperson’ approach. We have already briefly touched on the prominent position that some elite actors may hold in parts of ERS as potential sources of funding. Yet of course the ERS field may have many other potential interactions or avenues to work with influential actors, from prominent academics, developers of emerging technologies, politicians and civil servants, to name but a few. Engaging with high-visibility influential actors can be a natural point of entry for seeking to exert influence over ETR-mitigation measures. However, this is not just because they may be able to exert significant direct leverage on key policies, but also because they can be key public ‘spokespeople’, with the potential platform and considerable symbolic capital, that they are able to strengthen ETRs (either particular sources, or as a class) as credible threats to civilization, demanding and deserving response.

However, the relationship between influential elite actors and ETR can be complex. This is particularly the case where it comes to very visible ‘spokespersons’. To take one prominent if extreme, example, Elon Musk has been a proactive and early supporter of the field of ERS, providing both financial backing, and sitting on the scientific advisory boards of the Centre for the Study of Existential Risk and the Future of Life Institute, as well as investing heavily in technologies (e.g. batteries and electric vehicles) that could help reduce some global risks around climate and resources. At the same time, his idiosyncratic and eclectic approach to innovation and technology management has arguably stoked a competitive racing mentality in some ETR-relevant industries,⁶⁷ while

⁶⁷ For instance, Musk’s choices in the domain of AI development have received critique, with some suggesting that certain framings around the establishment of OpenAI may have spurred less cooperative dynamics around AI development, which may inadvertently erode the ability of different actors to cooperate around safe AI development [Benquo, 2017; Harth, 2021]. As such, this could increase the net risk of catastrophe, even when accounting for important direct work undertaken by the lab itself (both in terms of technical AI alignment, as well as in industry norm-setting work in domains such as release standards).

his approach to other global risks, such as the COVID-19 pandemic, has been dismissive [Walsh, 2021]. While the support and influence of particular types of elite actors may as such be valuable and welcome for raising awareness about- or making progress against some risks, it is nevertheless clear that at the very least the field of ERS needs to involve a diverse range of stakeholders, and be willing to distance itself from even its keenest supporters if they advocate for policies that research identifies as contributing to ETRs. At the same time, the community should be cautious about deploying such steps, and ensure these are not used to settle (non-ETR-related) political or personal disagreements amongst community members, or lead to interminable and unnecessary community splits in what should be, or might need to be, a broad-tent community working towards the shared purpose of mitigating ETRs.

Another actively used approach for pursuing influence in ERS is the ‘science-to-policy’ approach, which is more in line with the general ways in which other academic fields or ‘epistemic communities’ have sought to shape public debate and policymaker decisions.⁶⁸ The archetypical model of science-to-policy work is one where researchers, given the opportunity to address relevant stakeholders, and clear enough communication, are able to shape policies in evidence-based ways, to achieve desirable ends.⁶⁹ It is important for those in ERS who seek to pursue this model to improve pathways to linking scientific insight to policy change, which may require cultivating a distinct set of network and engagement skills [Rios Rojas, Richards & Rhodes 2021].

Simultaneously, a third and increasingly common approach in ERS is that of ‘academic activists’, perhaps best described as researchers working as academics-activists who are in the position of raising the

⁶⁸ For instance, on emerging epistemic communities’ work in shaping policy around military uses of AI, see [Belfield, 2020; Maas, 2019].

⁶⁹ The science-policy relationship is of course much more complex as described in Weiss [1979] and in the subsequent chapter of this book, by Sundaram and Currie [2021, this volume].

alarm about risks that have not been considered, or that have been too quickly dismissed, by those with responsibility for managing them. Of course, while this is one reason why it is valuable to analytically distinguish between ‘inside’ and ‘outside’ work, this should not be taken to suggest that these two approaches are at all mutually exclusive. Most efforts to achieve change invariably involve doing both kinds of work. This is legitimate, though such work should always ensure it considers and mitigates the risks of leaning strongly in one direction or the other. Whether conducting inside- or outside-work, it is important for scholars to find ways to build valuable networks of trust and influence (whether inside or outside of formal policy), while at the same time avoiding capture or co-option by other interests and meaningfully preserving academic freedom and the right to criticize.

How to get the balance right between these different forms of influence within ERS remains largely unresolved as a question for the community: as with other fields, there may be situations that bring out tensions between approaches pursuing inside- and outside-work. For instance, the partnerships that are required for effective, long-term inside working may unduly influence the output of research, thus rendering it a toothless or ineffective approach for those that seek to pursue outside working.

Conversely, outside work might produce research that is perceived as threatening or biased by decision-makers or by the public, thus spoiling opportunities for inside workers in the coalition. In particular, as in all academic fields, rushed or epistemically incautious outside work may make sweeping statements that reinforce framings or narratives about ERS issues. This could contribute to the unnecessary ‘ideological bundling’ of ERS issues [Colvin et al., 2018], leading to accelerated and counterproductive polarization of those issues along partisan lines. This might lock policymakers into ‘ideology traps’ [Buehler and Kessler, 2016] which could make progress or effective action, by any actor (whether inside or outside), impossible or

prohibitively costly. Of course, these are not challenges unique to ERS. But whether in this field or any other, it remains key to emphasize avenues by which good academic practice might mitigate the irreversible crystallization of such framings.

In this sense, outside work may be a higher-variance strategy, with more extreme outcomes: on the one hand, it potentially increases one's ability to identify and highlight important and neglected ETR issues, even (or especially) those that decision-makers prefer to ignore or downplay; on the other hand, it potentially exacerbates the downside costs of mistaken judgment, risking epistemic community coherence or credibility. There is as such of course a great deal more complexity in how ERS can and should approach its aims of mitigating existential risk than these simple dichotomies suggest. However, regardless of the means by which influence can be had, the onus is still largely on researchers to *do the influencing*, and that kind of action necessarily involves a degree of planning and strategy, implicit though it may sometimes be.

Question 6) How demographically diverse is the field of ERS, and what are the causes and consequences of this?

On the one hand, the question of diversity in ERS might appear a straightforward one to assess: given that the field is still small, it should not be difficult to gauge how homogeneous it is in terms of the background of scholars within it. However, in practice, as discussed at the start of this chapter, ERS is also a highly fluid field, at the nexus of many overlapping communities and academic fields, and many organizations hosting a wide range of visitors, collaborators and affiliates who may or may not identify (or be identified) as ERS scholars. This complicates any demographic 'census' of ERS.

The state of diversity in ERS

To our knowledge, there has only been one, relatively informal and non-comprehensive study in this domain, which only looked at the institutional subsection of the ERS field that is associated with and supported by the Effective Altruism movement [AnonymousEAForumAccount, 2020]. This assessment found that only 13-14% of researchers in this space were People of Colour (i.e. non-white), with almost half of ERS institutions surveyed (10/21) having only white members in their core staff, and a quarter (5/21) having only white members across all staff. It should be noted that there appears to be a significant difference between university-affiliated institutions (where 20% of staff are non-white) and those unaffiliated to universities (where only 3% are). There were also disparities among different roles, with more people of colour working in junior research and support roles and fewer in senior leadership positions. People of colour were, unsurprisingly, most represented among volunteer, part-time, freelance and internship positions where they made up 29% of all staff.⁷⁰

Other surveys indicate similar unbalances with regards to gender diversity, with 71% of respondents to a 2019 Effective Altruism survey reporting their gender as male, and 26.9% as female [Dullaghan, 2019b].⁷¹ This lack of gender balance, which could be even worse in other parts of the ERS community that have not kept data, is a challenge. It may be a problem for ERS not just for intrinsic normative reasons, but also if one takes a purely instrumental perspective on ETR-mitigation. For instance, Barnhart [2022] has argued that those concerned about existential risks of AI technology should advocate for the greater promotion of gender diversity amongst AI researchers, since polling suggests that women researchers are significantly more likely to think that issues of AI safety

⁷⁰ Indicating at the very least that there is no shortage of interest in and commitment to ERS, which we discuss further below.

⁷¹ As before, we re-emphasize here that the EA and ERS communities do not fully overlap, and so this is an imperfect proxy.

and AI governance research should be prioritized.⁷² In this way, the greater inclusion of women, both in decision-making roles in ETR-sensitive fields as well as in the ERS field itself, could be reliable and key interventions.

One area in which the field of ERS stands out as potentially more diverse than wider society is in relation to neurodiversity, mental health, and potentially other forms of disability. While no formal assessment of this in the community has been made, a reader survey of Slate Star Codex, a popular rationalist blog with some links to the ERS community, found that "[s]lightly less than a third of Slate Star Codex readers have anxiety; slightly more than a third have depression. Just under a fourth [...] are on the Autism spectrum (or believe themselves to be)". More tentatively there is also some evidence that the ERS community also has a greater than expected level of sexual and gender diversity,⁷³ the same survey found that "5.5% of [...] readers are transgender or genderqueer, [compared] to the 0.3% of American adults who identify as transgender."⁷⁴ Anecdotally we have found similar trends in the wider community. It is notable however that while a community likely gains substantially from having this diversity of ways of thinking and feeling about the world, this fact is often not mentioned, let alone celebrated. A greater understanding of neurodiversity and reflection on its benefits could be one way of opening up other conversations about the need for a plurality of voices and perspectives, as well as making the field more open and accessible to everyone.⁷⁵

⁷² Barnhart also cites evidence from polling, laboratory and field studies that indicates that risk perceptions vary by gender, with women demonstrating slightly higher risk perception both in general, but also in respect to emerging technologies in particular.

⁷³ While these are very different areas of diversity it is perhaps worth noting that GSD has a strong overlap with neurodiversity, although the reasons for this are unclear.

⁷⁴ Greer [2021], referring to data from [Slate Star Codex, 2020].

⁷⁵ This is discussed further in Wells-Jensen and Beard (forthcoming).

The causes of diminished diversity in ERS

Beyond the question of how to address any aversive symptoms of the existing demographic disparities in ERS, it is also key to investigate this at a deeper level: what could be the causes?

Of course, it could be that the diversity shortfall in ERS is simply reflective of academia as a whole. For comparison, statistics for the staff makeup of the University of Cambridge as a whole for 2018-2019 suggests that 13.8% of staff had a Black and Minority Ethnic (BAME) background; women made up 48.3% of all employees—but this split out to 63.1% of all assistant staff, and only 29.9% of research staff (below the Russell Group average of 42.8%) [University of Cambridge, 2020]. It is notable that even in STEM subjects (from which ERS draws some of its researchers and which are already home to a greater proportion of ethnic minority students and staff than non-STEM subjects), there may be limits: while a large-scale review of UK academia by The Royal Society [Joice and Tetlow, 2020:4-5] did not find a sizeable drop-off in ‘ethnic minority group STEM students continuing into postgraduate level STEM study’, it did significantly record “a drop-off of Black and mixed ethnicity STEM postgraduate students entering STEM academic employment at universities,” suggesting this remains a challenge for this subsequent, key academic career step. Given that there may be many research positions in ERS that are postdoctoral, dynamics such as these may therefore contribute to gradual demographic erosion, as young scholars attempt to move up in the tiers of academe and ERS.

If it draws from a stratum of academia that is already less diverse than earlier ones, it may be less surprising if ERS remains relatively homogeneous. It would be unreasonable to expect the relatively new and still small field of ERS to overhaul the entire academic system with its own attendant problems with inclusion. Moreover, lack of diversity is especially pronounced in the very fields where ERS originally found its

antecedents, such as philosophy [Botts et al., 2014, Schwitzgebel et al., 2021], or more recently the physical sciences and engineering [Joice and Tetlow, 2020].

Moreover, these challenges may be exacerbated by the fact that an academic career in ERS is not a secure one. In many respects, it is far less secure than a more traditional academic career in established fields, especially STEM [Currie, 2019] , because there are very few tenured positions or teaching opportunities. It is therefore not surprising that, after having ‘survived’ what can be a grueling experience during their PhD, few postdocs would want to risk their hard-earned qualifications on a somewhat risky career choice. Similar dynamics may be at play for segments of the community that encourage recent graduates with an interest in- or passion for ERS to pursue research career tracks in parallel to conventional academic careers. While valuable and important in some ways given the importance and urgency of the challenges at stake, there are features and misaligned incentives in this broad research pipeline [Aird, 2021a] which may mean that large number of interested applicants find their research ideas and career trajectories bottlenecked by the small number of high-profile ERS-focused organizations, which may not be able to directly accomodate all these applicants [Ngo, 2019], let alone provide them with reliable launch pads into (academic) career trajectories that could lead to more senior academic positions.

This of course pre-supposes that the problem lies in fewer people from underrepresented groups even applying to such institutions to begin with; data to support or refute this have not been analyzed. If this were the case, then a key challenge for ERS institutions would be to make themselves more attractive to a wider range of applicants. This can include things such as making jobs more flexible, advertising in a wider range of places, enforcing recruitment best practice, and providing

mentorship and coaching opportunities.⁷⁶ However, it can also mean taking steps such as showing willingness to engage with issues that are of greater concern to a greater diversity of underrepresented groups, such as the role of injustices in the creation of risk. However, even with all of these steps, if ERS is to continue being seen as a professionally precarious field then it is likely to attract only those candidates who can afford to take the risk of such precariousness and concrete measures are needed to ensure that this does not put off applicants who would only enrich the discipline.

A different (though not mutually exclusive) interpretation of the problem facing ERS might be that parts of the field, similar to other academic disciplines, are based around, or frequently draw upon, certain frameworks and ideas that have historically been associated with and/or misused in support of various forms of racial discrimination and oppression, misogyny, ableism, and imperial attitudes to society and nature. Much the same could be said for many disciplines.

The consequences of a lack of diversity for ERS

Beyond a challenge for talent, increasing the diversity of the field could have important and critical effects for helping ERS study ETRs effectively. As we discussed in Question 1, there are important implications for priority-setting in the field, and the current set of approaches, in isolation, could leave the field intellectually impoverished, with an overreliance on a single set of assumptions about the world that are potentially flawed, and certainly contested. At the same time, it should be noted that this kind of overreliance is neither universal nor static, and there are proactive efforts being made in many areas of ERS to diversify across many dimensions.

It would therefore be beneficial to the field of ERS, as one that aims to represent and safeguard the future of humanity as a whole, to pay

⁷⁶ There may be promising early initiatives in this space, such as the WANBAN network (now called Magnify Mentoring).

further attention to examining some of our core assumptions (and those of antecedent disciplines), and to further increase the diversity of disciplinary backgrounds engaged in the field, particularly those that have a more 'critical' understandings of their fields. Such work can draw on contemporary discourses on decolonization and diversification, and communicate this better to new entrants in the field. For instance, ERS could have an important role to play in critically questioning or even challenging overtly linear views of human progress, improvement, or perfection⁷⁷ or even developing entirely new approaches that are more able to embrace the whole of human experience. Although challenging to some, such discussions can be both creative and enlightening, especially as they enable the cross pollination of ideas as well as the opportunity to invite new people into discussions about ETR.

Working in ERS—working to avoid human extinction—does not mean uncritically holding up the status quo as the ideal [Mitchell and Chowdhury, 2020], or even as acceptable. To be sure, desiring continued human survival doesn't need to constitute an endorsement of how human societies are structured and ordered today, with their rife inequalities. Rather, this can be seen as a symmetrical, reciprocal relation: on the one hand, the avoidance of ETRs can serve as an important 'common cause' [Leech, 2018] and condition for lasting justice, as it may be necessary to ensure that both we as well as future generations will have the time and chance to correct those inequalities, and to recognize and repair (as far as possible) the deep injustices of the past and present [Ord, 2020].

Conversely and simultaneously, the re-ordering of those power structures is likely to play a significant part in ensuring that, if humanity does ultimately survive, it does so towards an actually desirable future, in which all can flourish rather than merely survive. So that the future will indeed look different from the past.

⁷⁷ Or at least clarify the potential for more complex cycles of post-disaster recovery in ways that are not necessarily linear [Baum et al., 2018; Belfield, forthcoming].

Question 7) How can ERS best reflect on and communicate about uncertainty in its field, as it continues to grow?

In light of the above questions, there is one additional question which we as ERS scholars can productively ask ourselves. All fields of enquiry involve uncertainty—after all there would be no point in research otherwise. However, for ERS, with all the challenges and issues we have here described, uncertainty is of particular, and perhaps obvious importance.

One particular challenge is that of, not simply *communicating uncertainty* (i.e. telling people our level of certainty in a given prediction or that a given strategy will be worthwhile or the ways in which a claim may be open to error and bias), but of communicating uncertainty *about uncertainty* (i.e. convincing people of the need to take action at all, while also being honest about how difficult it is to know what the problem is or how to solve it). Our best estimates or models when it comes to the nature of ETRs may well be the best of anyone's guesses, but they are sometimes still far from meeting conventional scientific standards of proof, and face a number of challenging critiques. In fact there is a great deal that differentiates the study of existential risks from the study of almost anything else. We must recognize that often (though not always) we will deal in the unprecedented, the unpredictable, the far off, and the extreme. We must take the perspective of humanity as a whole and understand our common future.⁷⁸

On the one hand then, many ERS scholars take their responsibilities of informing, advising and even influencing policy-makers, the public, and other stakeholders very seriously and thus need to develop tools for communicating to these audiences

⁷⁸ An aim which of course is made much easier in a context of diverse representation, as we explored in Questions 1 and 6.

appropriately—which is not necessarily ‘academically’. On the other hand, however the field does remain a scholarly/academic one (and according to some a *scientific* one), where, at least ideally, caution and transparency about the limitations of our methods and approaches are to be encouraged, and where concepts of falsifiability, objectivity, and impartiality are at least to be gestured at. Again, this is not an issue unique to ERS, and there is certainly much to be learned from other disciplines that have faced these challenges as well. Recognising that this is itself a key challenge, and the field is also evolving in this respect, moving from purely theoretical considerations to methodological ones: ERS scholars do—as in the Managing Extreme Technological Risks work that is the subject of this volume—adapt, refine and build on concepts and tools of ‘scientificity’ in ways that are appropriate to the field.⁷⁹

Regardless, this results in challenges of communicating (the uncertainties around) our research to several audiences, for different purposes. These purposes can include (1) communicating about ETRs in order to influence policy, (2) contributing to nuanced scientific understanding on the nature of ETRs and their management, and finally (3) to prompt reflection within the ERS community itself, about the questions we have raised heretofore.

One possible reaction here might be to propose differential communication strategies for each of these purposes and audiences. Following such an approach, the background research that consists of in-the-weeds and nuanced discussions about evidence, uncertainty and disagreement, would be mostly reserved for the academic debates, and ‘take home messages’ and high-level recommendations and strategies would be distilled and published separately for public and policymaker audiences, in a way that downplays some of the uncertainty and complexity for purposes of message clarity. However, there are risks to such a segmented communication strategy: not only would it feel

⁷⁹ See for example, the chapter by Sundaram and Currie [2022, in this volume].

somewhat intellectually patronizing, it might not work as intended—the underlying uncertainties are likely to be ‘found out’--and if so, this could easily backfire, by drawing into question our credibility, honesty and independence as researchers, no matter how benign our intentions.

Another approach is a much more intellectually honest one: making clear our uncertainty in as accessible a manner as we can, while nevertheless stressing the importance of tackling ETRs, when writing for either (1) policy or (2) the scientific field. For these cases, and these communities, it might be most appropriate to reflect openly on object-level uncertainties, while simultaneously conveying the underlying message that ETRs still need urgent attention and precautionary action: to stress that while we are uncertain (and may prove mistaken) about the specifics of a particular ETR, we can be confident about the aggregate importance and urgency of ETRs as an overarching human challenge.

However, that still leaves the additional question of the third purpose of communication: about how the ERS community itself should reflect on its own topic-prioritization (in terms of frameworks, disciplinary lenses, or funding), approaches, methods, theories-of-change and ways of pursuing impact, composition, and future trajectory. Of course this process of reflection about these questions, will be a continuation of existing conversations within the field, and will be a long-term endeavour. When we engage in these efforts, it will be important to recognize our uncertainties about these questions and their ‘right’ answers, while also understanding that that uncertainty is not sufficient reason to delay or avoid these conversations.

3. Conclusion

In this chapter, we have asked seven key questions of the ERS community; however, we have not attempted to provide definite answers to any of them. To do so would miss the point. These are hard problems that need careful consideration, and that consideration should start with an appreciation that there really are many different ways of answering these questions, each with advantages and drawbacks.

One of the reasons that we have chosen to address these questions now, is that they are rooted in our own lived experiences of being researchers in ERS. The Managing Extreme Technological Risk programme adopted an experimental approach to bringing together postdoctoral researchers from diverse disciplines to build understanding of several aspects of ERS and explicitly encouraged them to do so with connection to- and in collaboration with policy and scientific communities. About a year into the programme, its researchers entered into structured conversation with other CSER staff to consider some of the critiques of the ERS field that they were encountering, and how they might respond to them.⁸⁰ This chapter has therefore been informed by those initial discussions, along with later reflections on those questions with wider communities and with other emerging critiques.

These are particularly valuable conversations to have during the formation of a field like ERS, although they can at times provoke some discomfort, particularly where there are apparent areas of mismatch between ideals and reality (in regard to limited diversity in the field, for example). As an emerging and quite rapidly developing field of study ERS is – unsurprisingly – relatively small and has some ‘founder effects’. Nonetheless, there is already some diversity in relation to, for example, theories of change, areas of focus, and methodological preferences. This diversity is likely to continue to grow, although there may also be bounds to this if ERS is to remain a coherent field of study.

⁸⁰ This was the subject of part of a CSER-wide Away Day and formed the basis of some of the sessions in CSER’s CCCR2018 Conference.

Our hope must be that as the field grows larger and more established, this diversity and pluralism will increase, and that no single approach or theory will come to dominate or lead to greater homogenization. Crucially, there are also several ways in which ERS can benefit from reflecting on these questions, in a way that allows for more pluralism and diversity, both in the disciplinary backgrounds that can find a 'home' within the field, as well as in terms of who is attracted to the field, welcomed within it, and supported in their participation and contribution.

The only thing that would be worse than seeing this difference of opinions collapse within the field would be to find that people stop asking these questions at all because they become too controversial or difficult, or because the answers seem too uncertain. That would be a bad day indeed for this field, given the importance of what we work on for humanity as a whole.

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